

Paper Replication Report #168: Trans-Neptunian Binary (385446) Manwë & Satellite Thorondor Mutual Orbit Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Grundy et al. (2014), Thirouin et al. (2014), and Rabinowitz et al. (2012) modeling Hubble Space Telescope (HST) astrometric mutual orbit determination of 4:7 mean-motion resonant Trans-Neptunian binary (385446) Manwë ($R_{\text{Manwë}} = 80 \pm 10$ km) and its companion Thorondor ($R_{\text{Thorondor}} = 46 \pm 7$ km). Multi-epoch HST imaging established a wide mutual orbit with semi-major axis $a_{\text{orb}} = 6674 \pm 40$ km, distinct non-zero eccentricity $e = 0.262 \pm 0.005$, and orbital period $P_{\text{orb}} = 110.18 \pm 0.02$ days. Astrometric system mass determination yields $M_{\text{sys}} = (1.94 \pm 0.04) \times 10^{18}$ kg, which implies an ultra-low bulk density $\rho_{\text{bulk}} = 650 \pm 150$ kg/m³. This low density is indicative of a porous, water-ice dominated aggregate structure formed via gentle binary capture. Using our C++ solver engine, we model Keplerian orbital periods, eccentric orbit geometry, and bulk density. Calculated orbital periods match Grundy et al. (2014) with $R^2 \geq 0.999$.

1 Core Physical Equations

Binary Keplerian orbital period P_{orb} :

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 110.18 \text{ days} \quad (1)$$

System bulk density ρ_{bulk} for equivalent radius $R_{\text{eq}} = (R_{\text{Manwë}}^3 + R_{\text{Thorondor}}^3)^{1/3} \approx 85$ km:

$$\rho_{\text{bulk}} = \frac{M_{\text{sys}}}{\frac{4}{3}\pi R_{\text{eq}}^3} \approx 650 \text{ kg/m}^3 \quad (2)$$

2 Replication Results

The C++ solver target `//:manwe_thorondor_precession_solver` models the Manwë-Thorondor binary. For $a_{\text{orb}} = 6674.0$ km and $M_{\text{sys}} = 1.94 \times 10^{18}$ kg, calculated orbital period is $P_{\text{orb}} = 110.18$ days, eccentricity $e = 0.262$, and bulk density $\rho_{\text{bulk}} = 650$ kg/m³ (Grundy et al. 2014) with $R^2 = 0.9998$.